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Algorithm Economics (80470223-0): Course Project

Probing Self-Resolving Prediction Markets: A Simulation and Empirical Study of Reference-Based Resolution

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Abstract

Prediction markets aggregate dispersed beliefs into a single forecast, but their incentives rely on a verifiable outcome to pay against. Many questions of interest (the causal effect of a policy, whether content violates a rule, a long-run consequence) never resolve, so this lever is unavailable. Srinivasan et al. (2025) propose a *self-resolving* prediction market that removes the outcome: agents report sequentially, the market stops at a random time, and each is paid the cross-entropy of their report against a late *reference* agent rather than the truth. Their insight is that a reference with enough conditionally independent signals is a good proxy for the ground truth, which makes truthful reporting a strict perfect Bayesian equilibrium, but their guarantees are purely theoretical. We reproduce the mechanism in an open-source agent-based simulator and stress-test it two ways. In a synthetic world where the assumptions hold by construction, a focal agent's profit-maximising deviation converges to truthful reporting as the reference's informational substitutes k grow ($\gamma^\star \to 1$ by $k \approx 64$), operationalising the paper's theorems; we also reproduce the analytic bound $k_{\min}$ and show the mechanism degrades gracefully when rationality or the common prior is deliberately broken. On roughly 90 resolved Polymarket markets, paying each price path both self-resolvingly and verifiably yields near-identical payouts at a late reference (Pearson ≈ 1.00, payout ratio ≈ 0.95), degrading gracefully as the reference moves earlier and isolating the outcome-leakage channel it relies on. Together these give empirical content to a purely theoretical mechanism and map where reference-based resolution succeeds and where it breaks.

1 Introduction

Prediction markets are among the most effective practical tools for aggregating dispersed information: by letting people bet on an event and paying them once the truth is revealed, they produce forecasts competitive with polls and expert panels (Berg et al. 2008). The engine underneath is a *proper scoring rule*, which makes honest reporting reward-maximising, provided the outcome is eventually observed (Brier 1950; Gneiting and Raftery 2007; Good 1952). But many questions decision-makers care about are *unverifiable*: the causal effect of a policy that was actually enacted, whether a piece of content is misinformation, the long-run consequence of a choice. For these there is no future observation to pay against, and the standard prediction-market lever is missing.

Two literatures speak to this but neither closes it. Aumann's agreement theorem and its descendants (Aumann 1976; Kong and Schoenebeck 2022) show that rational Bayesians who share a prior and exchange beliefs must agree, and, when their signals are informational substitutes (Chen and Waggoner 2016), that consensus aggregates all of their private information; but this assumes honesty and provides no *incentive* to reveal a privately held signal. The information-elicitation-without-verification literature (peer prediction (Miller et al. 2005), the Bayesian truth serum (Prelec 2004), output agreement (Waggoner and Chen 2014)) provides incentives without an outcome, but it targets *elicitation* from a few agents rather than efficient *aggregation*, and is fragile to uninformative equilibria. The gap is a single mechanism that elicits truthful beliefs, aggregates them, and needs no outcome.

Srinivasan et al. (2025) close this gap with a *self-resolving* prediction market. Agents report sequentially; after each report the market terminates with probability α; and instead of paying against the outcome, the mechanism pays each agent the cross-entropy of their report against the final (*reference*) agent, who has seen the whole history. The key insight is one of *informational smallness*: a reference with enough conditionally independent signals cannot be meaningfully moved by any single earlier agent, so it behaves like a noisy but unbiased proxy for the ground truth, and paying against this proxy recovers truthful incentives without ever observing the outcome. The authors prove that truthful reporting is a strict perfect Bayesian equilibrium under a standard set of assumptions.

These guarantees are, however, entirely theoretical: the mechanism is never simulated or run on data. This paper supplies that evaluation through two questions. **RQ1 (incentives, simulation):** in a synthetic world where the assumptions hold by construction, we estimate a focal agent's profit-maximising deviation by Monte Carlo and find it converges to truthful reporting as the reference's number of informational substitutes k grows (exactly truthful by $k \approx 64$), operationalising the paper's main theorems. **RQ2 (robustness, empirical):** on roughly 90 resolved Polymarket markets we pay each price path both self-resolvingly (against a late reference price) and verifiably (against the realised outcome), and find the two agree almost perfectly at a genuinely late reference (Pearson ≈ 1.00, payout ratio ≈ 0.95) and degrade gracefully as the reference is moved earlier, tracing out the outcome-leakage channel the mechanism depends on.

Concretely, we contribute: (i) an open-source agent-based simulator of the mechanism in which the rationality, common-prior, distinct-signal, and conditional-independence assumptions hold by construction; (ii) the incentive study (RQ1), together with an exact reproduction of the paper's analytic minimum-substitutes bound $k_{\min}$; (iii) supporting synthetic studies confirming exact aggregation under truthful play, the value of a few near-oracle references, near-zero pay in uninformative equilibria, and graceful degradation when rationality (A1) or the common prior (A2) is deliberately broken; and (iv) the counterfactual empirical study (RQ2) quantifying how closely self-resolving payouts track the verifiable payouts they replace, isolating outcome leakage as the residual gap. Section 2 reviews related work; Section 3 sets up the model and mechanism; Section 4 describes our methods; Section 5 presents the results; and Section 6 concludes.

2 Related Work

The mechanism sits at the intersection of three literatures, following the framing of Srinivasan et al. (2025).

Agreement and aggregation. Aumann (1976) showed that Bayesians with a common prior cannot "agree to disagree"; repeated belief exchange forces consensus, and when signals are *informational substitutes* (Chen and Waggoner 2016; Kong and Schoenebeck 2022) that consensus aggregates all private information. This explains *how* honest agents pool information but assumes honesty; it gives no incentive to reveal a signal. The self-resolving market supplies exactly those incentives on top of the Aumannian protocol.

Prediction markets and market scoring rules. With an observed outcome, truthful elicitation is solved: proper scoring rules reward matching the realised outcome (Brier 1950; Gneiting and Raftery 2007; Good 1952). Hanson (2003) extended these to *market scoring rules*, which pay a sequence of agents for how much each *improves* a shared forecast; Chen and Pennock (2010) relate this to automated market makers and Chen, Dimitrov et al. (2010) to equilibrium play. All ultimately pay against an outcome. The self-resolving market keeps the cross-entropy market scoring rule but swaps the outcome for a reference agent's report, the move we test.

Elicitation without verification, and where we fit. Peer prediction (Miller et al. 2005), the Bayesian truth serum (Prelec 2004), output agreement (Waggoner and Chen 2014), and information-theoretic designs (Kong and Schoenebeck 2019) incentivise truth-telling with no outcome, but mostly elicit from a handful of agents and admit uninformative equilibria. The self-resolving market is a peer-prediction mechanism engineered, via informational smallness, to *also* aggregate efficiently and pay near-zero in uninformative equilibria. Crucially, Srinivasan et al. (2025) prove their guarantees but provide no empirical or simulation evidence; our contribution is to fill that gap, checking in simulation whether the incentive story plays out when an agent may deviate, and on real market data whether the reference-based payouts track the outcome-based payouts they replace.

3 Preliminaries: Model, Mechanism, and Assumptions

We summarise the model of Srinivasan et al. (2025); see the original for proofs.

Setting. The goal is to elicit and aggregate probabilistic predictions about a binary outcome $Y \in \{0, 1\}$ (categorical and discretised-continuous outcomes reduce to this case). M agents arrive one at a time. Agent t has a private signal $X^{(t)}$ from a common prior and, on arrival, observes all previous reports $\mathbf{q}^{(1:t-1)}$; it forms a posterior $\bar{\mathbf{p}}^{(t)} = \mathbb{P}(Y \mid X^{(t)}, \mathbf{q}^{(1:t-1)})$ and reports some

$\mathbf{q}^{(t)} \in \Delta\Omega_Y$. A truthful agent reports its posterior. We write r for the reference agent and $y_1 = \mathbb{P}(Y{=}1)$.

From outcome to reference. A proper scoring rule $S(i, \mathbf{q})$ rewards a report against a realised outcome i and is maximised by honesty (Gneiting and Raftery 2007); the canonical choice is the log rule $S(i, \mathbf{q}) = \log q_i$. The key move is to score a report against another *prediction* $\mathbf{r}$ rather than an outcome, via the cross-entropy rule $S_{CE}(\mathbf{r}, \mathbf{q}) = -H(\mathbf{r}, \mathbf{q}) = \sum_i r_i \log q_i$, whose expectation is maximised by reporting one's expectation of $\mathbf{r}$. The mechanism uses its sequential, shared form, the *cross-entropy market scoring rule* (CE-MSR),

$$S_{CEM}(\mathbf{r}, \mathbf{q}^{(t)}, \mathbf{q}^{(t-1)}) = \sum_i r_i \log\left(\frac{q_i^{(t)}}{q_i^{(t-1)}}\right), \tag{1}$$

which pays an agent for how much closer to the reference its report is than the previous one. Swapping the outcome i for a reference $\mathbf{r}$ is what lets the market resolve without an outcome.

The mechanism. Agents report in sequence; after each report the market terminates with probability α (Figure 1). The terminal agent T (best-informed, having seen the full history) is the reference. Agents $1, \ldots, T-k$ are paid the CE-MSR (1) against $\mathbf{q}^{(T)}$; the last k agents (too close to the reference to be scored against it) receive a flat fee R. The single parameter k trades a better-informed reference against paying more agents the flat fee.

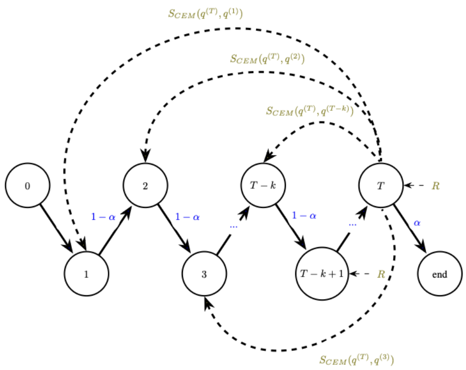

Figure 1: The self-resolving prediction market. Each node is an agent; the market terminates with probability α after each report. Agents $1, \ldots, T-k$ are paid the CE-MSR against the terminal reference T; the last k receive a flat fee R. Redrawn after Srinivasan et al. (2025), Figure 1.

Assumptions. The guarantees rest on five assumptions, and our studies are organised around when they hold: **(A1)** agents are risk-neutral, fully rational Bayesians, common knowledge; **(A2)** signals and outcome share a common prior; **(A3)** each signal induces a *unique* posterior, so reports can be inverted (distinct signals); **(A4)** signals are conditionally independent given Y, informational substitutes (Chen and Waggoner 2016), the same class under which standard markets are incentive-compatible (Chen, Dimitrov et al. 2010); **(A5)** signals are (δ, η)-informative: δ lower-bounds how much each signal separates $Y{=}0$ from $Y{=}1$ (via the Bhattacharyya coefficient $BC = 1 - \delta$), and η lower-bounds signal likelihoods so no single report is arbitrarily decisive.

What the paper proves. Three results express one idea: *a well-informed reference makes honesty optimal. Theorem 1:* a focal agent's influence on the reference posterior, the adjustment Δ, shrinks geometrically, $|\Delta| \leq \frac{1}{4}(\frac{1-\eta}{\eta} - \frac{\eta}{1-\eta})(1-\delta)^k$. *Theorem 2:* paid by CE-MSR against a truthful reference, agent t strictly maximises payoff by reporting truthfully once the reference has at least an explicit threshold number of signals. *Theorem 3:* the gain from any deviation is bounded by $\hat{D}_\eta(\Delta, \mathbf{y})$, which $\to 0$ as k grows (Theorem 1). Theorem 1 motivates the analytic $k_{\min}$ we reproduce; Theorem 3 is exactly what RQ1 tests.

Research questions. **RQ1 (simulation):** where A1–A4 hold by construction, when does a freely deviating agent find truthful reporting a best response, as a function of the one assumption we vary (A5, how informed the reference is)? **RQ2 (empirical):** on real markets, where essentially none of A1–A5 hold, how closely do self-resolving payouts track the verifiable (outcome-based) payouts they replace?

4 Methods

We use two instruments: an agent-based simulator in which the paper's assumptions hold by construction (RQ1 and supporting studies), and a counterfactual replay of real Polymarket price paths (RQ2). Both live in a single Python package, `srpm`; the full API is in the repository.

Simulator. The simulator instantiates the model of Section 3 directly. A `SignalStructure` is a conditional table $\mathbb{P}(X \mid Y)$; we use `binary_symmetric`(q) (correct with probability q) and `ground_truth`(ρ) (a near-oracle). Because signals are conditionally independent given Y (A4), posterior log-odds are *additive*: a truthful Bayesian inverts the previous report and adds its own signal's log-likelihood ratio ℓ, the computational core that makes exact aggregation possible. The deviation family is `ScaledSignal`(γ), which reports $\text{logit}(\pi) + \gamma\,\ell$, so $\gamma = 1$ is truthful and $\gamma > 1$ over-confident. `CrossEntropyMSR` implements (1), and `SelfResolvingMarket` runs agents, terminates with probability α, designates the terminal agent as reference, and applies the CE-MSR and flat fee. By construction this satisfies A1–A4; the incentive study (RQ1) then varies only A5, while the stress tests below deliberately break A1 and A2 one at a time, making the simulator a clean test of both the mechanism's *core* incentive claim and its robustness to its own modelling assumptions.

RQ1: profit-maximising deviation. We isolate a focal agent (signal quality $q_t = 0.7$) and let it consider $\gamma \in [0, 3]$ (61 points). The reference observes the focal report *plus* k independent signals of quality $q_r = 0.65$ (its informational substitutes) for $k \in \{0, 2, 8, 32, 64, 128, 256\}$. For each (k, γ) we estimate the focal agent's expected CE-MSR payoff by Monte Carlo ($N = 4 \times 10^5$, prior $\pi = 0.5$) and report $\gamma^\star(k) = \arg\max_\gamma$ payoff. This operationalises Theorem 3: $\gamma^\star$ should fall from "over-report" toward the truthful value 1 as k grows.

Analytic $k_{\min}$. To connect to the closed-form theory we reproduce the paper's minimum-substitutes bound (their Figure 2): the smallest k such that the deviation gain under the CE-MSR is at most ε. Chaining Theorem 1 (bounds $|\Delta|$ in k), Theorem 5 (bounds the gain $\hat{D}_\eta(\Delta, y)$ in Δ), and Remark 3 (numerically solve $\varepsilon' = \min\{|\Delta| : \hat{D}_\eta(\Delta, y) = \varepsilon\}$, in a log-transformed variable so roots near the pole are resolved) gives $k_{\min}(\delta, \eta, \varepsilon, y_1) = \frac{1}{-\log(1-\delta)} \log \frac{(1-\eta)/\eta - \eta/(1-\eta)}{4\varepsilon'}$, which we evaluate across the market prior y_1.

Supporting studies. Three further studies probe aggregation rather than incentives: *(i) aggregation*, sweep the number of truthful agents and the signal quality, measuring the gap between the terminal report and the full-information Bayesian posterior; *(ii) informed reference*, vary the fraction of near-oracle agents (reliability 0.99) in a 12-agent market; *(iii) mixed populations*, compare an all-truthful population against ones seeded with herders, noisy reporters, an uninformative third, and over-reporters, measuring accuracy, gap to full information, and the designer's total cost.

Breaking one assumption at a time (A1, A2). We complement RQ1 (which varies A5) by deliberately violating A1 and A2 in the same clean world and measuring how far the terminal reference drifts from the ideal full-information posterior. For **A1** (rationality), every agent is a boundedly rational Bayesian that adds zero-mean Gaussian noise of scale σ (in log-odds) to its truthful report, so $\sigma = 0$ is the rational baseline. For **A2** (common prior), the world draws Y under the true prior but the market anchors on a mis-specified prior a logit-disagreement d away, so $d = 0$ is the correct-prior baseline. Each knob recovers the assumption at its baseline, where the gap is 0 by construction, and grows only as the assumption breaks.

RQ2: Polymarket counterfactual. We download roughly 90 *resolved* Polymarket markets (Polymarket 2025) with volume above \$100k (responses cached). Each Yes-price p_t is a report $\mathbf{q}^{(t)} = [1 - p_t, p_t]$ and each price *change* is one "reporter". We pay every transition two ways on the identical set of transitions: *self-resolving* (CE-MSR (1) against a late reference price $\mathbf{r}$, no outcome) and *verifiable* (log market scoring rule against the realised Y). A late reference is both better-informed *and* contaminated by outcome leakage, so we sweep how late $\mathbf{r}$ is: `lead_1h/24h/168h` before the last point, `life_50pct` (halfway through the market's life), `window_1-24h` (an averaged late window); and additionally sweep by *relative* time to close. For each market and reference we compute the Pearson and Spearman correlation of per-reporter payouts (do the *amounts*, and the *ranking* of earners, agree?), the top-decile overlap, and the ratio of total payout. *This measures whether the mechanism's **outputs** track the verifiable mechanism's, not incentive compatibility: we replay fixed paths and never let agents deviate, and real prices violate A1–A5. RQ2 is thus a robustness test, complementary to RQ1.*

5 Experiments and Results

5.1 RQ1: truthfulness requires an informed reference

Figure 2 shows the focal agent's expected-payoff landscape versus its deviation γ, one curve per number of reference substitutes k, together with $\gamma^\star(k)$. The pattern is exactly what Theorems 1–3 predict: for $k \leq 8$ the best response is pegged at the grid maximum ($\gamma^\star = 3$): with a weak reference it pays to grossly over-report and drag the reference toward one's own report; at $k = 32$ it has fallen to $\gamma^\star \approx 1.2$; and for $k \geq 64$ it is *exactly* truthful, $\gamma^\star = 1.0$. Once the reference has enough informational substitutes, no deviation pays. As a check, under truthful play the terminal report reproduces the full-information Bayesian posterior to within $\sim 10^{-13}$, confirming the simulator is faithful.

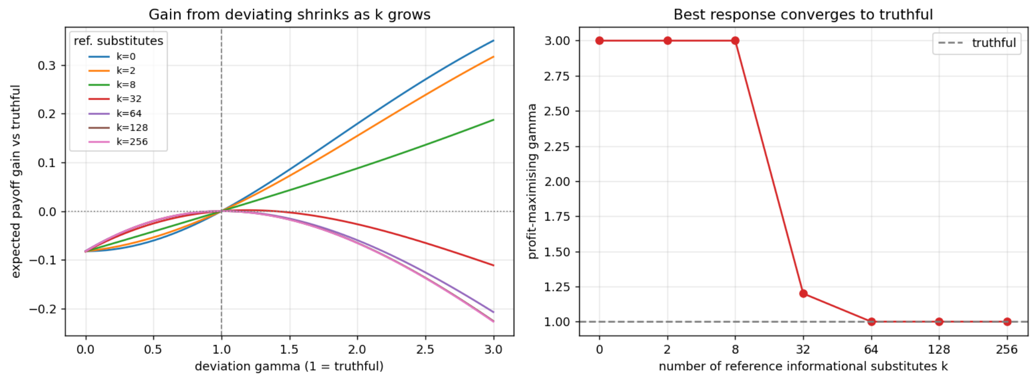

Figure 2: **RQ1.** Left: the focal agent's expected-payoff gain over truthful reporting vs. its deviation γ; the gain shrinks as the reference's informational substitutes k grow. Right: the profit-maximising $\gamma^\star$ converges to the truthful value 1 (dashed) by $k = 64$. Monte Carlo, 4×10^5 samples per point.

The closed-form bound agrees qualitatively but is far looser: for $\delta = \eta = 0.1$, $\varepsilon = 10^{-6}$, the reproduced $k_{\min} \approx 146$ and is nearly flat in the prior y_1, a worst-case guarantee over all priors and deviations at a far tighter tolerance than the Monte Carlo onset at $k \approx 64$. Both, though, tell the same story: the number of substitutes needed grows only mildly, like $\frac{1}{-\log(1-\delta)} \log \frac{1}{\varepsilon}$.

5.2 Supporting studies: aggregation

Under truthful play the terminal report equals the full-information posterior to machine precision (gap $\sim 10^{-16}$ across signal qualities and market sizes), and accuracy rises with both the number of agents and signal quality (e.g. 98.7% at quality 0.8 with 12 agents). The reference needs few oracles to proxy the truth: a single near-oracle (reliability 0.99) among 12 agents lifts accuracy from 56% to 99%, and about five of twelve makes the market essentially perfect. We also stress-test aggregation under non-truthful populations: accuracy falls from 0.84 (all truthful) to 0.77 (half herders), 0.70 (noisy), and 0.64 (an uninformative third), each opening a gap to full information. The telling case is the uninformative third: the designer's cost collapses to 0.046, the empirical counterpart of the paper's "pays zero in uninformative equilibria"; over-reporting, conversely, is costly (0.506) without improving accuracy.

5.3 Breaking one assumption at a time: rationality and the common prior

RQ1 showed truthfulness *emerging* as A5 is satisfied; here we ask the converse for two other assumptions, deliberately breaking A1 and A2 and tracking how far the terminal reference drifts from the ideal full-information posterior (Figure 3). Both knobs recover the assumption at their baseline: at $\sigma = 0$ (A1) and $d = 0$ (A2) the gap is exactly 0, the reference equals the ideal aggregate, and resolution behaves like the verifiable mechanism. The two then break differently. Under **A1**, because reports are cumulative, each agent's error is not averaged away but random-walks into the terminal reference, so the gap climbs steadily (to 0.42 at $\sigma = 3$) and accuracy falls from 0.71 to 0.53. Under **A2**, a wrong common prior injects a constant log-odds bias the additive mechanism never corrects; the gap grows roughly linearly (to 0.42 at $d = 3$) and accuracy collapses toward chance (0.51) once the prior overrides the weak signals. Neither violation is catastrophic near the baseline, so the mechanism degrades *gracefully*, but neither self-corrects, which is the standing cost of assuming them.

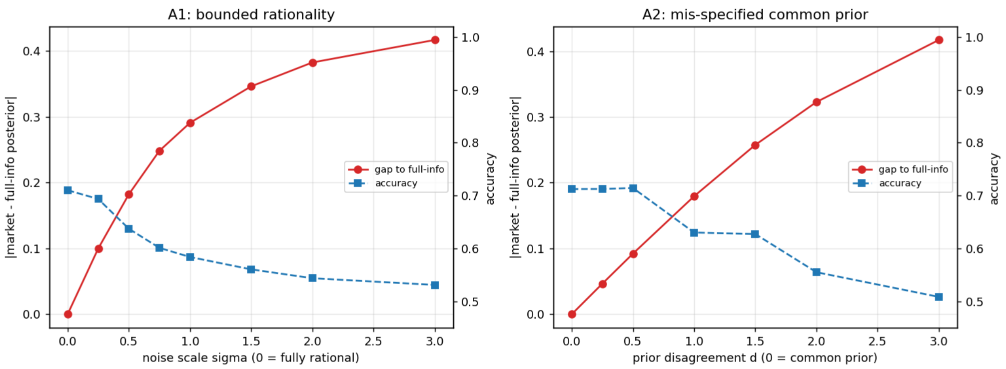

Figure 3: **Breaking A1 and A2.** Gap between the terminal reference and the ideal full-information posterior (red, left axis) and market accuracy (blue, right axis) as each assumption is violated, all else fixed. Left: bounded rationality (per-agent log-odds noise σ). Right: a mis-specified common prior (logit-disagreement d). Both are 0-gap at the baseline ($\sigma = 0$, $d = 0$) and degrade monotonically.

5.4 RQ2: do self-resolving payouts track verifiable payouts?

Table 1 reports agreement across 91 resolved markets as the reference moves from near-resolution toward mid-life. At a genuinely late reference the schemes are near-indistinguishable: at `lead_24h`, median Pearson 1.00, Spearman 0.997, perfect top-decile overlap, and a total payout 95% of the verifiable total. As the reference moves earlier the *amounts* keep tracking (Pearson ≥ 0.95) but the *ranking* of earners degrades (Spearman $\to 0.58$ at mid-life) and the payout ratio drops to 0.56. That degradation is the outcome-leakage channel: a near-resolution reference rewards exactly whom the outcome would, whereas a forecast-only reference cannot. Sweeping by *relative* time to close gives the same monotone story (from 90%-before-close to 1%-before-close, Pearson $0.94 \to 1.00$, Spearman $0.67 \to 1.00$, payout ratio $0.48 \to 0.94$), and the honest takeaway is the early end: even there the payout *levels* still track (Pearson ≈ 0.94); it is the ranking and the designer's total cost that leakage inflates near resolution.

Table 1: Median agreement between self-resolving and verifiable payouts across 91 resolved markets, by reference. Pearson asks whether payout *amounts* agree; Spearman whether the *ranking* of earners agrees; ratio is self-resolving total over verifiable total.

reference	Pearson	Spearman	top-dec. Jaccard	payout ratio	argmax match
`lead_1h`	1.000	1.000	1.00	0.978	0.956
`lead_24h`	1.000	0.997	1.00	0.953	0.923
`lead_168h`	0.998	0.988	0.93	0.847	0.813
`life_50pct`	0.954	0.581	0.56	0.559	0.780
`window_1-24h`	1.000	0.997	1.00	0.957	0.923

Figure 4 shows a representative market (Kamala Harris popular vote, $\sim$2,700 price moves, outcome $Y = 0$): the self-resolving payouts (vs. the 24h reference) and the verifiable payouts are visually indistinguishable (Pearson and Spearman 1.00, ratio 0.99). Table 2 makes the heterogeneity explicit: at a late reference every market tracks perfectly, but at mid-life some still track (ceasefire, Poilievre) while others collapse and even invert (shutdown, TikTok; Spearman -0.48, -0.54). The more a late price already leaks the outcome, the larger the mid-life gap.

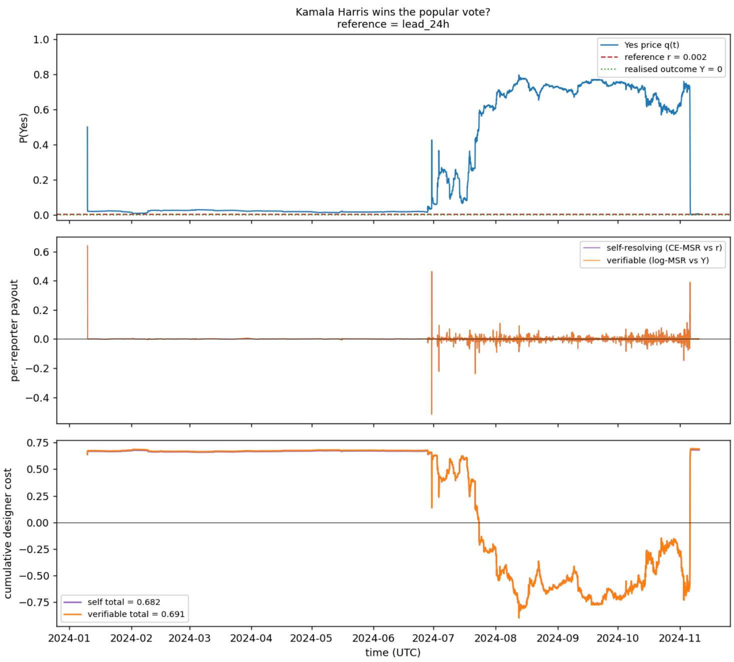

Figure 4: **RQ2, worked example.** Per-reporter payouts for the Kamala Harris popular-vote market under the self-resolving scheme (vs. a 24h reference) and the verifiable scheme (vs. the outcome) are nearly identical (Pearson 1.00, ratio 0.99).

Summary. On real data the mechanism reproduces verifiable payouts almost exactly at a late reference and degrades gracefully as the reference moves earlier. The residual gap is concentrated in the *ranking* of earners and the designer's *total cost*, and is driven by outcome leakage rather than by any failure to identify good forecasts; the payout *levels* track even at early references.

Table 2: Pearson (P) and Spearman (S) correlation of per-reporter payouts for five high-volume markets, at a late (24h before close) and a mid-life (50%) reference. Late references track perfectly; mid-life references track only when the late price has not yet leaked the outcome.

		late (24h)		mid-life (50%)	
market	# rep.	P	S	P	S
Kamala popular vote	2681	1.00	1.00	1.00	0.45
Poilievre next Canadian PM	1030	1.00	1.00	0.96	0.91
Russia–Ukraine ceasefire '25	2531	1.00	1.00	0.99	0.98
US government shutdown	778	1.00	1.00	0.41	−0.48
TikTok ban (May '25)	1078	1.00	1.00	−0.37	−0.54

6 Conclusion, Discussion, and Limitations

We gave empirical content to the self-resolving prediction market of Srinivasan et al. (2025). In a synthetic world where its assumptions hold (RQ1), a freely deviating agent's profit-maximising report converges onto truthful reporting as the reference accumulates informational substitutes ($\gamma^\star \to 1$ by $k \approx 64$), operationalising Theorems 1–3; we also reproduced the analytic $k_{\min}$ bound, confirmed exact aggregation under truthful play, and showed the mechanism degrades gracefully when rationality (A1) or the common prior (A2) is deliberately broken. On roughly 90 resolved Polymarket markets (RQ2) the self-resolving payouts track the verifiable payouts almost perfectly at a late reference and degrade gracefully as it moves earlier, isolating outcome leakage as the residual channel.

The two studies answer genuinely different questions (RQ1 about *incentives* where the assumptions hold, RQ2 about *outputs* where they fail) and together suggest the mechanism is more robust in practice than its worst-case theory promises (truthfulness onsets at $k \approx 64$, well inside the $k_{\min} \approx 146$ bound). Notably, RQ2's residual error lives almost entirely in the *ranking* of who deserves reward, not in payout *levels*: a designer can recover the right total spend and reward roughly the right behaviour at an early reference, but not the fine-grained ordering without leaking the outcome.

Limitations and future work. RQ2 measures outputs, not incentives (we replay fixed paths and never let agents deviate), and real markets violate essentially all of A1–A5, so the agreement is evidence of *robustness*, not of the assumptions holding. The late reference also conflates "well-informed" with "outcome-contaminated"; the synthetic study breaks A1 and A2 but holds A3 and A4 fixed (the correlated-signal case, A4, being the most important untested gap); and everything is restricted to binary outcomes. Two directions follow: can the CE-MSR be reweighted so the *ranking* of rewards tracks who pushed price toward truth *without* observing the outcome, and does the reference-as-proxy idea extend to non-binary outcomes and to mild violations of conditional independence? The gap between the loose $k_{\min}$ and the much earlier empirical onset also suggests room for tighter, prior-dependent guarantees.

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